# IoT Management: A Framework for Responsible Artificial Intelligence Water Consumption

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## INTRODUCTION

Sustainable Development Goal (SDG) 9 focuses on Industry, Innovation, and Infrastructure. It aims to build resilient infrastructure, promote sustainable industrialization, and foster innovation, which are key aspects of economic growth and sustainable development. SDG 9 also supports Least Developed Countries (LDCs) by investing in advanced technologies and sustainable infrastructure. Furthermore, it supports the development of modern solutions that make work easier, improve productivity, and lessen environmental damage. SDG 9 is important because it helps create employment opportunities, generate stable incomes, and promote technological advancement that benefits societies and economies [1].

Water is a vital and most important resource for human life, agriculture, and industrial activities. However, water scarcity is a problem that can affect communities, agriculture, and industries. The cause of water scarcity is that humans have used natural water resources by creating structures such as dams, wells, and irrigation systems to support daily life. Moreover, the freshwater today is increasingly overused causing some lakes, rivers, and ground water sources to dry up [2]. As water scarcity continues to be a global concern, effective water management and responsible water consumption have become essential for ensuring the sustainable use of water. The system should involve the collection, distribution, utilization, and conservation of water resources [3].

In response to these challenges, modern technologies have created new opportunities for improving water management. Technologies like Internet of Things (IoT) and Artificial Intelligence (AI) can help collect, monitor, and analyze water consumption more efficiently. The role of IoT in water management is real-time monitoring and data collection.[4] AI can help water conservation by resource tracking, and reducing waste to support better resource management. These innovations have the potential to promote water consumption and contribute to the objectives of SDG 9 by encouraging innovation and sustainable development. This study explores the use of IoT and AI technologies in water management to support responsible water consumption and contribute to sustainable resource management.

## PROPOSED SOLUTION

### Overview of the Proposed System.

This study proposes a Water-Aware AI Workload Scheduler, an IoT-based system that treats water as a factor in deciding when artificial intelligence (AI) jobs are run. AI workloads can have an indirect water footprint because they generate heat on the servers that process them, while some data centers rely on water-intensive cooling systems to dissipate that heat. The amount of water associated with the same amount of computing can also vary with location and operating conditions, making the timing of workloads relevant to water efficiency [4], [5]. Conventional scheduling objectives commonly emphasize performance, energy, or cost; this framework adds water use as an additional scheduling consideration [4].

The proposed system begins by gathering environmental and water-related data through IoT sensors and uses these measurements to determine an appropriate time and operating mode for AI jobs, with the goal of reducing estimated water consumption without compromising timeliness [4], [6]. The four stages of the framework are: (1) Sensing, (2) Scoring, (3) Decision, and (4) Evaluation. The overarching goal is to develop and test a prototype of an IoT-based scheduler that minimizes estimated water use of AI workloads. Specifically, the work will focus on creating a sensor node to measure ambient temperature, humidity, water reservoir level, and power consumption of the computing device; creating a water stress score based on the measurements; implementing a scheduler that runs, throttles, or defers an AI job based on the measured water stress score and the deadline of the corresponding job; and comparing the proposed scheduler with a first-in, first-out (FIFO) baseline in terms of estimated water consumed, job delay, and missed deadlines.

### System Architecture and Hardware Components.

The system consists of a number of layers, including a sensing layer, an edge layer, a compute layer, a control layer, and a visualization layer. The sensing layer has a DHT22 sensor for temperature and humidity, an ultrasonic or float sensor for the water reservoir level, and an

INA219 sensor for current and power consumption of the computing device. The sensors are connected to an ESP32 microcontroller that acts as the edge node and publishes the sensor data to the MQTT every few seconds. The sensor data can be used to monitor water in real time in a water-management system, and MQTT is a lightweight publish/subscribe protocol for IoT and constrained environments [6] and [7]. The ESP32 is used for Wi-Fi connectivity and interfaces compatible with IoT sensor nodes; the INA219 is used to measure the bus voltage, current, and power, with an I2C/SMBus interface [9, 10]. The prototype is based on the "mini data center" built by Raspberry Pi. It has the Python scheduler and job queue and processes sensor data, performs small AI applications such as image classification or small language model inference. This information is displayed in a dashboard via Grafana or Node-RED or Streamlit which include live readings, water stress score and estimated water footprint. The connections between water and cooling also are demonstrated in the prototype; the use of a "relay-controlled" fan or small pump for the cooling system is optional.

**Water Stress Score and Scheduling Logic.**

The summary of the current water-related operating conditions is displayed through a Water Stress Score (WSS) (0-1) which is used by the scheduler. The normalized value of each sensor reading is calculated and combined using the weights shown below. The selected variables are based on the relationship between environmental conditions, cooling requirements, computing load and water use in data center cooling systems [4, 5].

$$WSS = w1(Temp) + w2(1 - Humidity) + w3(1 - Reservoir\ Level) + w4(Load)$$

It is desirable to have a higher temperature, as there is a potential for more heat demand; a lower air moisture percentage may have a negative effect on evaporative cooling; lower reservoir level indicates less water available; and a higher computing load can add heat to be dissipated [5]. After a large sample of weights is obtained, the weights are tuned experimentally or fitted by a simple regression model. This study proposes the WSS equation, not a pre-existing standardized formula. These AI job roles are then divided into Urgent, Flexible, or Deferrable. Real time jobs are always executed right away as it is an urgent job. If the score is low or medium, the flexible jobs, such as the batch inference, will be run. A deferrable job is a job that is executed only during times when the score is not a maximum score such as model training or retraining. When the score is small, then a job will be running up at full speed. If it is moderate,

then the work is slowed down to a certain point by decreasing the number of threads or size of the batch. If the high, then postpone the job. A deferred job is always executed when its time is almost up to keep it from going to sleep, so that jobs are not abandoned but deferred.

**Life Cycle Water Footprint Estimation and Evaluation.**

The water footprint is determined for the small prototype and expressed using Water Usage Effectiveness (WUE), measured as liters of water per kWh of energy. WUE is an established data-center sustainability metric that relates site water use to IT energy consumption [5], [8]. The energy used by each job is calculated from the INA219 power readings over time, and the estimated water use is calculated as energy multiplied by WUE. The WUE value is treated as dynamic in the proposed model because water efficiency can vary with operating and weather conditions; research on AI water footprints shows that runtime water efficiency can vary across time and location [4], while data-center cooling guidance shows that temperature, humidity, and cooling strategy affect water use [5]. Onsite cooling water can also be considered alongside offsite water associated with electricity generation when estimating the broader water footprint [4].

The same set of AI jobs is tested on the FIFO baseline and the water-aware scheduler, both at various hours of the day and with water temperature recordings and/or simulation. The two are compared based on the estimated amount of liters of water per job batch, the average job delay, number of missed deadlines and total energy consumption. The anticipated outcome is a clear picturesque trade-off between the number of water saved and the delay added, which will show if the water saved is a result of only the timing. A small physical demonstration using a reservoir, flow sensor and fan can be added to demonstrate the real flow of water, and stating that the results are estimates from a scaled model.

**Alignment with the Sustainable Development Goals.**

The proposed solution directly supports SDG 9 (Industry, Innovation and Infrastructure) – Target 9.4, which calls for upgrading infrastructure and industries to be sustainable in terms of their capacity to be more efficient at the use of resources and more resilient to the use of

environmentally friendly technologies [1]. The study combines digital innovation with IoT sensing and a scheduling framework to prioritize water demand to optimize resource efficiency. The proposed solution is also aligned to SDG 6 (Clean Water and Sanitation), particularly with the goal of improving water use efficiency across all water use sectors and addressing water scarcity (6.4) [11]. The solution is also associated with SDG 12 (Responsible Consumption and Production), specifically Target 12.2: By 2025, achieve sustainable management and efficient use of natural resources [12]. Further, a reduction of cooling demand will lead to reduced associated energy consumption, thereby also contributing to the SDG 13 goal on Climate Action (renewed [5], [13]).

## CHALLENGES AND LIMITATIONS

In the process of developing and producing new technologies, the designers eventually encounter limitations and problems due to various factors. With the correct strategies that tackle the root of the problems, these challenges can be mitigated. The researchers identified challenges and theorized strategies that addressed the accessibility and feasibility of the proposed prototype, assessing whether it is sustainable and aligns with SDG 9 (Industry, Innovation, and Infrastructure).

### Challenge 1: Infrastructure Limitations

The AI workload scheduler runs on a Raspberry Pi, a small prototype, which has no water cooling, making the water figure an estimated value by multiplying the measured energy by an assumed WUE value [9], rather than an accurate measurement. Since WUE changes with many variables, such as weather and location, the estimated liters may differ from what an actual data center could consume. To address this, the dashboard will display the assumed WUE values alongside the results, in line with calls for clearer reporting of AI's water footprint [5], showing the resource efficiency gains that align with Target 9.4 in a transparent and credible manner.

### Challenge 2: Security Concerns

Since the ESP32 publishes sensor data to the scheduler through an MQTT broker, an unsecured broker is a risk [4]. If the broker accepts unauthenticated access, fake readings could be injected, which might cause the scheduler to defer jobs unnecessarily. To address this, the

ESP32 node will connect to the broker with authentication credentials, a username and password. The scheduler will also reject the readings outside plausible ranges before calculating the WSS, supporting the reliable infrastructure in Target 9.1.

**Challenge 3: Trade-off Between Saving Water and Job Delays**

In the proposed solution, deferring AI jobs away from high-stress periods can reduce estimated water use, but it risks delaying results and missing deadlines. Water-efficient timing may also conflict with carbon efficiency, since cooler night hours save cooling water while solar power is cleanest during the day [5]. To address this, the scheduler applies the "Unfollow the Sun" idea by shifting Flexible and Deferrable jobs away from hot and high-stress periods [5]. Urgent jobs always run immediately, and deferred jobs are forced to run before their deadlines, so deferred work is not abandoned. This reflects SDG 9's goal of sustainable infrastructure, balancing of water savings and timely results.

## CONCLUSION

Water scarcity remains a global concern and significant challenge that affects the sustainability of water resources and highlights the need for more efficient water management practices. This study proposes a Water-aware AI Workload Scheduler that combines IoT technologies and Artificial Intelligence (AI) to monitor environmental conditions and optimize better water usage and management. The proposed system utilizes IoT sensors and AI-based decision-making to monitor environmental and water-related conditions, helping improve water efficiency and reduce estimated water consumption through real-time monitoring and intelligent scheduling. Furthermore, the device demonstrates how innovative the technologies can support water use and improves its water management efficiently. In addition, the proposed solution supports the objectives of SDG 9(Industry, Innovation and Infrastructure), SDG 6 (Clean Water and Sanitation), and SDG 12 (Responsible Consumption and Production) by promoting innovation, improving water-use efficiency, and encouraging the responsible utilization of natural resources. Overall, this study highlights the potential solutions for water scarcity to address the challenges and support long-term sustainability.

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